Security, Access Control, and Backup Policy

1. Role-Based Access Control (RBAC)

We implemented RBAC in our Streamlit dashboard to ensure data privacy:

- **Admin:** Full access to all dashboards and all department data.
- **Analyst:** Read-only access to the Overview and Monthly Trends.
- **Department Head:** Restricted access. If a Department Head logs in, the system dynamically updates the SQL query to only show data for their specific department (e.g., Engineering).

2. Data Masking

Student IDs are masked in the visual dashboards (e.g., showing only STU-2024-XXX) to protect Personally Identifiable Information (PII).

3. Backup & Recovery Strategy

- **Full Backup:** A complete `mysqldump` of the `library_dw` database will be executed every Sunday at 2:00 AM.
- **Incremental Backup:** Binary logs will be backed up daily to capture changes.
- **Recovery Plan:** In the event of a failure, the latest full backup will be restored, followed by replaying the binary logs up to the point of failure to ensure zero data loss.